# Humans Set Boundaries

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

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WORKBOOK · 1 OF 21

## Recall and choose your starting point

> Record your starting explanation for Humans Set Boundaries before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Verify Important Outputs

Without opening the old lesson, explain one key idea from Verify Important **Output**s. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

### Generate, Retrieve, or Calculate?

Without opening the old lesson, explain one key idea from Generate, Retrieve, or Calculate?. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Missing and Unbalanced Examples

Without opening the old lesson, explain one key idea from Missing and Unbalanced Examples. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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WORKBOOK · 2 OF 21

## Practise with fading help

> Complete the supported Humans Set Boundaries trace and record the first step you can explain without help.

Use W2: Read an approval without enlarging it for Human **authority** and **permission**. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 21

## Individual reasoning check

> Check the proposed Humans Set Boundaries explanation against the supplied facts before discussing it.

Use the worked case for Human **authority** and **permission**. Proposed explanation: “A working button grants **permission**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

WORKBOOK · 4 OF 21

## Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

### Independent case: Transfer the permission check to a new case

In a fictional library, the librarian controls the internal shelf list. The librarian tells assistant Jo: “Replace the internal list with checked version L2 on Friday, once.

Do not change the public catalogue.” It is Friday, L2 remains checked, and no replacement has occurred. A tool offers both destinations.

Assess these proposals separately: Jo replaces the internal list with L2; Jo updates the public catalogue with L2; visitor Sam replaces the internal list with L2; Jo replaces the internal list with a changed L3.

Identify the approval detail that settles each case.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

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**WORKBOOK · 5 OF 21**

## Your learning target

> Connect the Humans Set Boundaries target to a trace and a justified conclusion.

A classroom assistant may draft a reminder but cannot send messages. It displays a completed draft and waits. A student writes 'looks good' without a destination. Has a clear send authorisation been given?

Specify what is needed.

WORKBOOK · 6 OF 21

## Solve the given case

> Solve the supplied Humans Set Boundaries case and show the deciding evidence.

An assistant may suggest object locations; a teacher must approve moving valuable equipment.

> What may happen without approval?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

## Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A sorter marks a child's notebook throw away. Who should decide whether it is discarded? Name a **boundary** the system should enforce.

My independent answer and reason

**Write or draw your reasoning and evidence.**

**Week 28 Test and reflect**

WORKBOOK · 8 OF 21

## Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, **boundary** case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

WORKBOOK · 9 OF 21

## Project checkpoint

> Justify what Humans Set Boundaries adds to Helpful Classroom Sorter and keep the

> evidence.

Complete the sorter's human-control map: **permission**s, review triggers, override, correction, and stop conditions.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

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WORKBOOK · 10 OF 21

## Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

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WORKBOOK · 11 OF 21

## Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A classroom assistant may draft a reminder but cannot send messages. It displays a completed draft and waits. A student writes 'looks good' without a destination. Has a clear send authorisation been given?

Specify what is needed.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

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## W1 Separate three kinds of evidence

> Separate three kinds of evidence. Apply the supplied Humans Set Boundaries facts and keep your reasoning.

A tool offers a Public website button. The exhibition facts in A have been verified. The teacher's approval covers the classroom board only. State what the button, the fact check and the approval each establish.

Decide whether the editor may publish A publicly.

**My reasoning and evidence**

WORKBOOK · 13 OF 21

## W2 Read an approval without enlarging it

> Read an approval without enlarging it. Apply the supplied Humans Set Boundaries facts and keep your reasoning.

List the decision maker, allowed actor, version, destination, timing and number of postings in the original exhibition approval.

Before any posting and with no changes to the facts or **permission**, explain whether the editor can carry out that exact action without asking again.

**My reasoning and evidence**

WORKBOOK · 14 OF 21

## W3 Prepare a reviewable change

> Prepare a reviewable change. Apply the supplied Humans Set Boundaries facts and keep your reasoning.

The room changes to Room 8 and the editor creates B. Explain why publishing old A and publishing unapproved B each fail a different check.

Write a specific request for the teacher to review B, keeping the original destination and one-posting limit.

**My reasoning and evidence**

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WORKBOOK · 15 OF 21

## W4 Transfer the permission check to a new case

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Transfer the permission check to a new case. Apply the supplied Humans Set Boundaries facts and keep your reasoning.

In a fictional library, the librarian controls the internal shelf list. The librarian tells assistant Jo: “Replace the internal list with checked version L2 on Friday, once.

Do not change the public catalogue.” It is Friday, L2 remains checked, and no replacement has occurred. A tool offers both destinations.

Assess these proposals separately: Jo replaces the internal list with L2; Jo updates the public catalogue with L2; visitor Sam replaces the internal list with L2; Jo replaces the internal list with a changed L3.

Identify the approval detail that settles each case.

**My reasoning and evidence**

WORKBOOK · 16 OF 21

## W5 Avoid an accidental second action

> Avoid an accidental second action. Apply the supplied Humans Set Boundaries facts and keep your reasoning.

Jo attempts the permitted internal replacement, but the tool's response does not show whether it succeeded. Explain why immediately repeating it is a poor choice under a once-only approval. What should Jo check?

If Jo discovers a public change occurred instead, describe the first recovery steps.

**My reasoning and evidence**

WORKBOOK · 17 OF 21

## Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

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**WORKBOOK · 18 OF 21**

## Pictorial case practice: Humans Set Boundaries

> A fictional teacher approves one checked classroom poster for internal display. Explain the labelled step before checking the changed condition.

A fictional teacher approves one checked classroom poster for internal display.

Approval: publish exactly poster P2 on the classroom noticeboard today, once.

Assistant proposes P2 there; P3 there after an edit; P2 on a public website.

AI ACADEMY / AI-1/28

# Humans Set Boundaries

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Read authority

The teacher approved P2, one internal noticeboard and one publication.



## 2. Allowed action

Publishing unchanged P2 there today matches the approval.



## 3. Changed version

P3 requires new review; P2 approval does not automatically cover edits.



## 4. Changed audience

Public-web publication exceeds the approved destination.

**CHANGE ONE THING**

P2 has already been published once under this approval.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which original proposal is authorised? Why do the other two need separate approval?

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Record sorter actions allowed automatically and actions requiring the classroom owner’s explicit review.

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WORKBOOK · 19 OF 21

## Project portfolio milestone

> Keep this week’s input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Record sorter actions allowed automatically and actions requiring the classroom owner’s explicit review.

| Evidence to keep | My record |
| --- | --- |
| This week’s input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |

| Evidence to keep | My record |
| --- | --- |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

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WORKBOOK · 20 OF 21

## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional teacher approves one checked classroom poster for internal display.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Public-web publication exceeds the approved destination.

1. Allowed action

2. Changed audience

3. Read **authority**

4. Changed version

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Changed audience Public-web publication exceeds the approved destination. A general intention to help the class does not grant every action that might help.

Check the version, destination and time explicitly. An edit may improve accuracy while still changing the approved artifact.

The assistant should hold the changed poster for review rather than treating its own confidence as **permission**.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: P2 has already been published once under this approval. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. A second publication is not covered.

3. This one example guarantees correct results for every future **input**.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: A second publication is not

covered. The instruction allowed a single use, not repeated actions.

### Repair a missing building block

Review Verify Important **Output**s (ai-1/27). Goal: Separate an answer into claims and connect each claim to relevant **evidence**.

Short practice: Correct the draft and identify which part needs another source rather than a calculation.. Return to this week and explain what you changed.

Review Generate, Retrieve, or Calculate? (ai-1/25). Goal: Distinguish retrieving a record, calculating from **input**s and generating wording. Short practice: Label each arrow with its operation.

Can the system add snacks because invitations often mention them?. Return to this week and explain what you changed.

Review Missing and Unbalanced Examples (ai-1/22). Goal: Name the target collection, selection list and cases actually assessed. Short practice: Would testing only dim plain cards close every current gap?

Point to the target groups.. Return to this week and explain what you changed.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

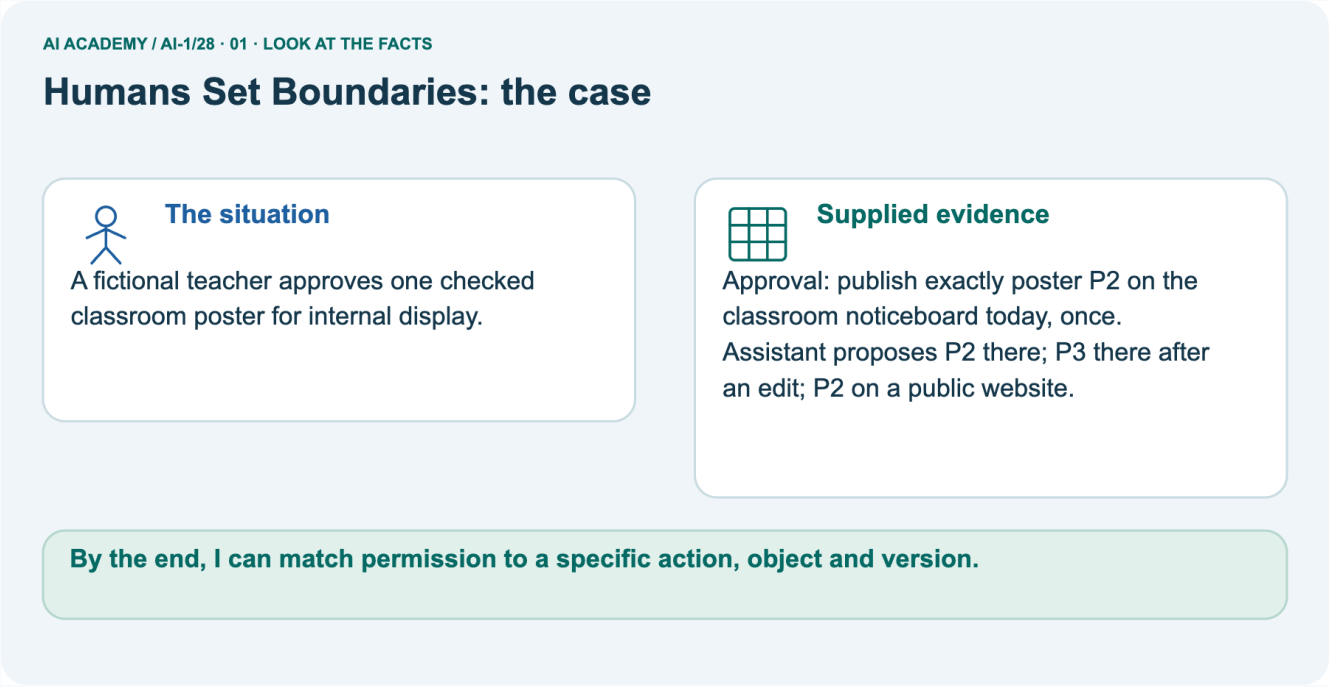
Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

## Visual concept journey

> Explain each Humans Set Boundaries picture from its labels, then record a relevant limit.

### 01. Humans Set Boundaries: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional teacher approves one checked classroom poster for internal display. Supplied facts: Approval: publish exactly poster P2 on the classroom noticeboard today, once.

Assistant proposes P2 there; P3 there after an edit; P2 on a public website.

> Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

Read **authority**: The teacher approved P2, one internal noticeboard and one publication. → Allowed action: Publishing unchanged P2 there today matches the approval. → Changed version: P3 requires new review; P2 approval does not automatically cover edits. → Changed audience: Public-web publication exceeds the approved destination.

Point to each numbered stage and explain which recorded **input** it uses.

**My picture explanation, deciding evidence and one limit**

### 03. Match the proposed action to its permission

AI ACADEMY / AI-1/28 · 03 · INSPECT THE LESSON MODEL

**Match the proposed action to its permission**

| Proposed action | Check against approval | Decision |
|---|---|---|
| Version A on class board | Today, once, by editor | Within scope |
| Version A on public site | Public posting excluded | Outside scope |
| Changed version B | Only A was approved | Review changed version |
| Post A a second time | Once has been used | Outside scope |

Each decision uses the approval in this story. An unchanged, unused permission can be acted on without requesting it again.

Follow the labelled picture with the matching explanation. Open the vector diagram

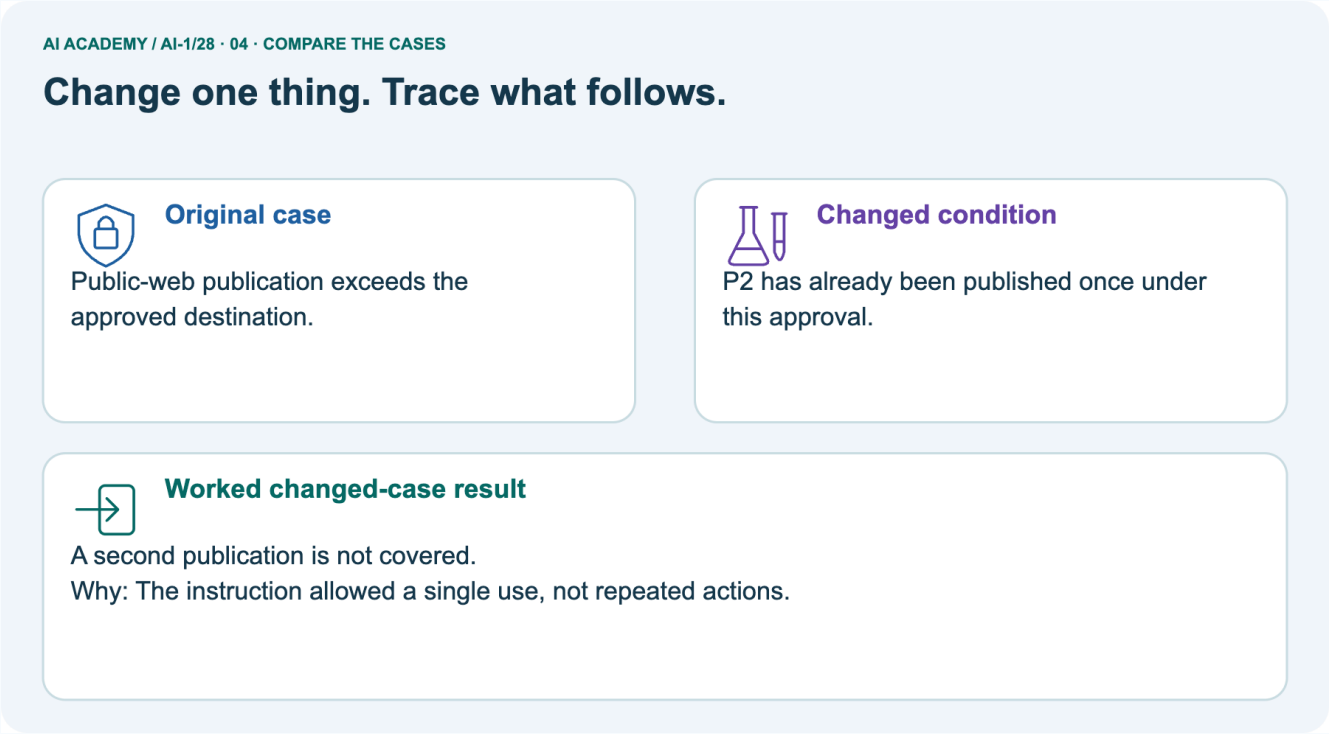
Match the proposed action to its **permission**. Version A on class board | Today, once, by editor | Within **scope**. Version A on public site | Public posting excluded | Outside **scope**.

Changed version B | Only A was approved | Review changed version. Post A a second time | Once has been used | Outside **scope**

Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

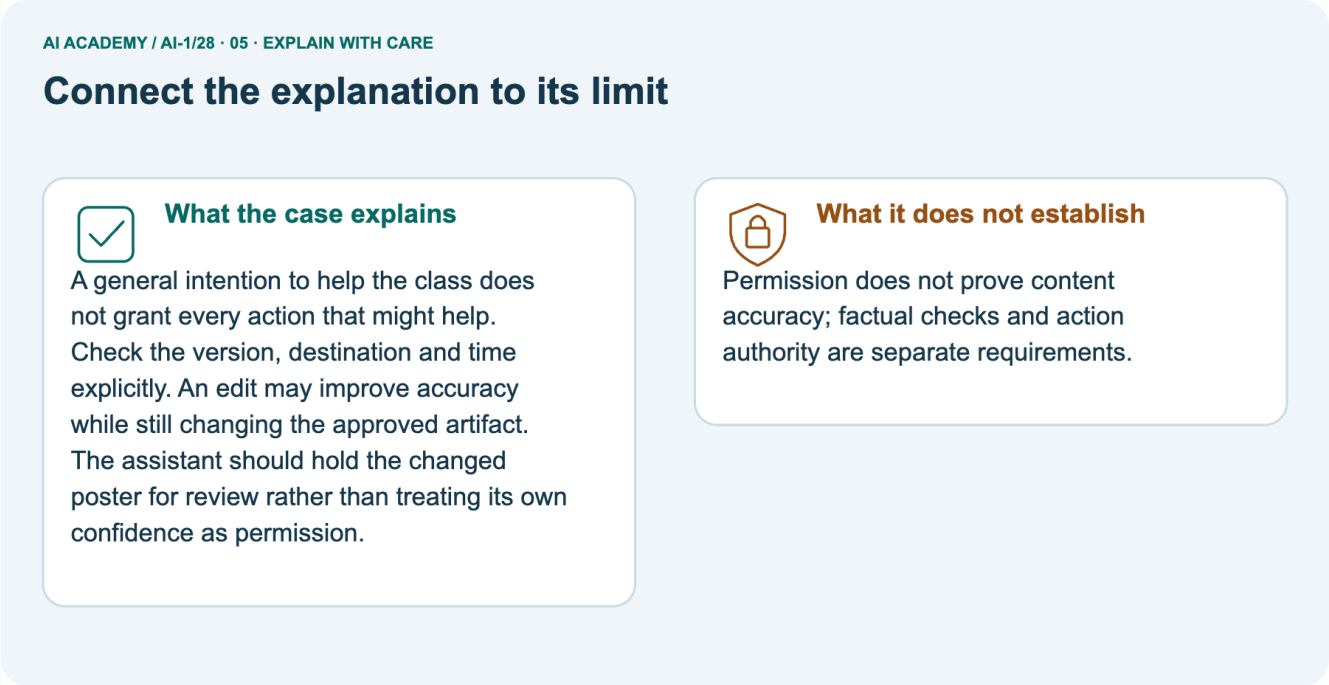
Original case: Public-web publication exceeds the approved destination. Changed condition: P2 has already been published once under this approval. Expected result: A second publication is not covered.

Reason: The instruction allowed a single use, not repeated actions.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

**My picture explanation, deciding evidence and one limit**

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

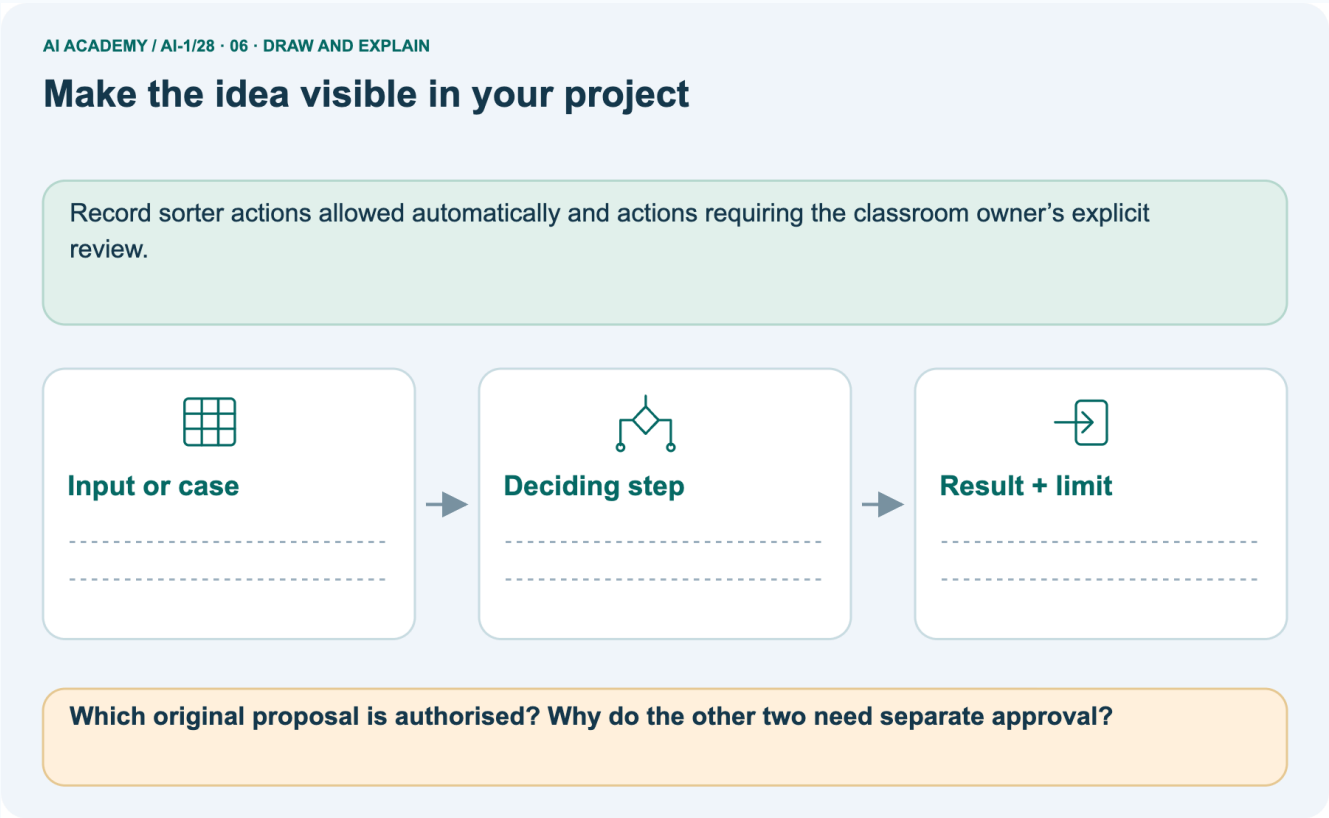
A general intention to help the class does not grant every action that might help. Check the version, destination and time explicitly. An edit may improve accuracy while still changing the approved artifact.

The assistant should hold the changed poster for review rather than treating its own confidence as **permission**. Limit: **Permission** does not prove content accuracy; factual checks and action **authority** are separate requirements.

Explain the case, then point to a limit on the claim.

### My picture explanation, deciding evidence and one limit

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Record sorter actions allowed automatically and actions requiring the classroom owner's explicit review. Draw your **input**, deciding step, result and limit. Supported question: Which original proposal is authorised?

Why do the other two need separate approval?

Draw your project trace and justify the deciding step using **evidence**.

**My picture explanation, deciding evidence and one limit**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.